Research Implementation Summary

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Main research contribution

I adapted studied conjugate-gradient-type formulations to the variants used in my work, implemented the numerical algorithms in Python, and developed MPI-parallel versions. I personally produced the numerical efficiency comparisons and formulated the conclusions reported in my May 2026 seminar. The starting theoretical manuscript and established CG methods are background sources, not claimed as my inventions.

Programming and MPI implementation

Algorithm implementation	Expressed the selected iterative formulations as executable numerical procedures; investigated two CG-type implementations.
MPI parallelization	Distributed matrix blocks on 2-D Cartesian process grids; developed row/column communicators, local products, and collective reductions.
Efficiency analysis	Adapted the computation/communication cost-model framework, programmed the formulas, generated efficiency comparisons for selected matrix sizes and process topologies, and drew my own conclusion.
Other numerical work	Implemented experimental MPI cross approximation and supporting inverse-problem notebooks, including CGNR and low-rank preconditioning tests.

May 2026 seminar and individual conclusion

My Russian-language seminar compared two conjugate-gradient-type formulations for ill-posed linear inverse problems. It discussed numerical differences, computation and MPI communication costs, and parallel-efficiency estimates. My comparison found that the two formulations had similar estimated parallel efficiency for the investigated configurations.

Results versus direct runtime measurements

The efficiency curves in the supplied E.py and E2.py scripts are my own computational results, obtained by evaluating parameterized cost formulas for serial and parallel times. They are **model-based results**, not direct wall-clock measurements of running the MPI programs. This distinction concerns the experimental measurement method and does not diminish my authorship of the implementation, adaptation, plots, or conclusions.

Evidence and current verification scope

The GitHub repository contains the Python/MPI programs, comparison scripts, original seminar slides, and notebook results. The included notebook files were re-executed while packaging. MPI files were not executed in that packaging environment, so hardware timing and distributed-runtime verification are not claimed by this summary. An accompanying personal-contributions file describes the individual role in more detail.

Project: numerical-inverse-problems-hpc | Source code: see README.md and CONTRIBUTIONS.md